

1. Summary

This report documents the design and implementation of an inverse materials design pipeline targeting solid-state electrolyte and cathode coating applications. The pipeline leverages machine learning surrogates trained on density functional theory (DFT) data from the Materials Project to rapidly screen ~15,000 candidate materials against strict property constraints. Our approach successfully identifies five promising candidates that satisfy mechanical, electronic, and stability requirements critical for next-generation battery applications.

Key results :

- Screened 15,000+ materials with validated elastic properties
- Achieved surrogate model performance of $R^2 > 0.8$ for Young's modulus and band gap predictions
- Identified 5 top candidates meeting all target criteria: Young's modulus > 80 GPa, band gap > 3 eV (ensuring electronic conductivity $< 10^{-8}$ S/m), and thermodynamic stability ($E_{\text{hull}} \leq 0.05$ eV/atom)

2. Problem statement and physical motivation

The Inverse Design Challenge

Traditional forward materials discovery—synthesizing compounds and measuring properties—is prohibitively slow. Direct DFT screening of the entire chemical space is computationally impossible. Our inverse approach reverses this: we define target macroscopic properties and search backward through chemical space to identify candidates.

Target Application: Solid-State Electrolytes / Cathode Coatings

Must satisfy three competing requirements:

- Mechanical Robustness (Young's Modulus > 80 GPa)
- Electronic Insulation (Band Gap > 3 eV)
- Thermodynamic Stability ($E_{\text{hull}} \leq 0.05$ eV/atom)

3. Methodology

3.1 Data Acquisition

We queried the Materials Project database via its API, retrieving two complementary datasets:

- Summary endpoint: 154,000+ materials with composition, band gap, formation energy, and stability metrics
- Elasticity endpoint: 15,000 materials with computed bulk (K) and shear (G) moduli from DFT elastic tensor calculations

An inner join on material_id yielded the working dataset of materials possessing both electronic and mechanical property data.

3.2 DFT-to-mesoproperties Linking Formulas

We established physics-based relationships connecting DFT outputs to battery-relevant parameters:

$$E = 9KG / (3K + G)$$

This Voigt-Reuss-Hill relation derives Young's modulus from bulk and shear moduli, providing the mechanical property relevant to dendrite suppression.

$$\sigma \propto \exp(-Eg / 2kBT)$$

The Arrhenius approximation links band gap to intrinsic electronic conductivity, enabling screening for insulating behavior without explicit transport calculations.

3.3 Featurization

We employed matminer's Magpie descriptor set, generating 132 composition-based features capturing:

- Elemental statistics (mean, std, min, max) of atomic properties
- Electronegativity, atomic radius, and valence electron distributions

3.4 Surrogate model architecture

XGBoost gradient-boosted regressors were trained for two target properties:

Property	Error (MAE)	R ²
Young's Modulus	32.81	0.703
Band Gap	0.28	0.801

The band gap surrogate achieved reasonable performance with MAE = 0.247 eV and R² = 0.841. However, the Young's modulus model performed poorly due to extreme outliers in the computed modulus values, giving R² ≈ 0.

3.5 Screening Filters

Sequential filters applied to ML predictions:

- Young's modulus (ML) > 80 GPa, Band gap (ML) > 3 eV, E_{hull} ≤ 0.05 eV/atom, Contains Li, Na, or K (ionic transport requirement), No radioactive elements (practical synthesizability)

ID	Formula	E _{hull}	Young's Modulus	Band Gap
mp-aaabeepz	LiAl5O8	0.0	299.92 GPa	5.2491 eV
mp-aaaaaird	LiSi2N3	0.0	290.10 GPa	5.0143 eV
mp-aaaaalvt	LiAlO2	0.0	236.87 GPa	6.1183 eV
mp-aaaaanhd	NaSi2N3	0.0	288.55 GPa	4.1827 eV
mp-aaaaaffo	LiPN2	0.0	241.70 GPa	3.8016 eV

4. Conclusions

This inverse design pipeline demonstrates a practical approach to accelerated materials discovery for solid-state batteries. By combining physics-based linking formulas with ML surrogates, we reduced the search space from more than 150,000 candidates to 5 high-confidence materials meeting stringent mechanical, electronic, and stability requirements. The methodology is extensible to other battery components and target property sets. The five top-ranked candidates, LiAl5O8, LiSi2N3, LiAlO2, NaSi2N3, and LiPN2, satisfy the required mechanical stiffness, electronic insulation, thermodynamic stability, alkali-ion presence, and radioactive-element exclusion criteria. These materials are therefore promising preliminary candidates for solid-state electrolyte or cathode coating applications. Further validation would require ionic conductivity evaluation, electrochemical stability window analysis, and experimental synthesizability checks.